

Sensor Noise Analysis in Event-Based Measurements of Laser-Reflected Surface Vibrations

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Abstract

This work investigates the signal characteristics and noise behavior of an event-based vision sensor under frequency-modulated interferometric stimulation. Using a region-of-interest (ROI) analysis framework, event activity was quantified across multiple modulation frequencies ranging from 75 Hz to 115 Hz. Three complementary metrics were developed to assess sensor performance: spatial noise, capturing pixel-wise variability across the sensor array; temporal consistency, quantifying the regularity of event timing within regions of interest (ROIs); and signal-to-noise ratio (SNR), measuring the strength of modulation signals relative to baseline noise following the framework of Moeys et al. Spatial analysis revealed distinct regions of heightened variability, suggesting localized sensor non-uniformities or bias imbalances. Temporal analysis indicated that while most active ROIs maintained consistent event timing, several exhibited irregular temporal firing patterns associated with noise-dominated regions. SNR trends showed that frequencies near 90–95 Hz demonstrated an optimal operational band where stimulus-driven responses dominate noise fluctuations. Together, these findings demonstrate the viability of event-based sensors for interferometric signal detection and establish a robust framework for quantifying spatiotemporal coherence and noise performance in neuromorphic sensing applications.

I. Introduction

Laser interferometry is a powerful optical technique that extracts quantitative information from the interference of coherent light waves. It has long been used for high-precision measurements of displacement, vibration, and surface motion in scientific and engineering applications [1, 2, 3, 4]. One common implementation is the laser vibrometer, which measures surface vibrations by comparing the laser light reflected from the target surface with a reference beam. When these two beams recombine at the sensor, their interference pattern encodes the phase variations caused by the surface motion, enabling accurate estimation of vibration amplitude and frequency.

Among existing optical vibrometry techniques, interferometric methods such as Michelson interferometry are widely used due to their ability to measure nanometer-scale displacements with high precision [5, 6]. These systems rely on the interference between a reference and a reflected beam, where the resulting fringe patterns encode surface motion (Fig. 1). However, traditional frame-based implementations of interferometric vibrometers are constrained by the speed and cost of high-frame-rate cameras. Capturing high-frequency vibrations requires rapid frame acquisition, which generates massive data volumes, increases latency, and introduces computational bottlenecks in post-processing [8, 9].

To overcome these limitations, we explore the use of event-based cameras for optical interferometry. Event-based sensors record changes in logarithmic intensity asynchronously at the pixel level, providing microsecond temporal precision, high dynamic range, and dramatically reduced data throughput compared to conventional imaging sensors [7, 10, 12]. These properties make event cameras particularly suited for measuring dynamic optical signals, where rapid variations in interference fringes occur due to surface vibrations.

Previous research has explored the usage of event cameras for optical vibration sensing, most notably in the Event2Audio framework developed by Cai et al [7]. In their work, event-based cameras were employed to recover audio signals underwater, and the primary focus was on achieving high-quality reconstruction of sound as a proof-of-concept for event-based vibrometry. In contrast, our work does not aim to reconstruct audio signals, but instead, we shift focus toward the characterization of sensor noise within event-based interferometric measurements. Understanding the nature of this noise is critical for assessing the limits of sensitivity and the

reliability of phase reconstruction in event-based optical systems.

In this work, we specifically aim to identify and characterize sensor-level noise in event-camera measurements of laser-reflected surface vibrations. Optical noise is assumed negligible or indirectly linked to acoustically induced fluctuations, allowing us to isolate sensor-driven noise contributions. The analysis focuses on pixels most affected by acoustic vibrations—those located in regions of high interference contrast. We quantitatively evaluate the impact of sensor noise on event accuracy relative to expected modulation cycles, spatial consistency of pixel responses, and the overall inferred signal-to-noise ratio (SNR). The specific project goals are as follows:

1. Characterize sensor noise mechanisms expressed in event data.
2. Quantify the effect of sensor noise on event accuracy and pixel behavior.
3. Infer SNR and identify pixels with abnormal activity.
4. Provide practical recommendations for reducing sensor noise in future experiments.

Overall, this work bridges event-based sensing and optical interferometry by experimentally evaluating both the vibrational response and the underlying sensor noise characteristics, establishing a foundation for more reliable, high-speed event-based vibrometry in future optical measurement systems.

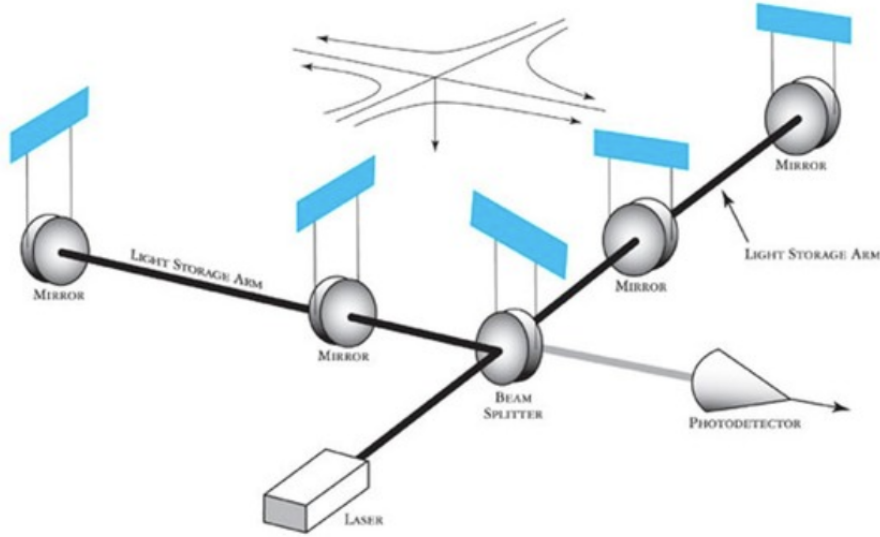


Figure 1: *Standard Michelson interferometry setup*

II. Theoretical Background

The basic theory of the Michelson interferometer is discussed in this section. Essentially, the subtle motion of reflective surfaces can be measured with precision using optical interferometry, which converts nanoscale displacements into measurable interference patterns of coherent light. Specifically, when two beams—one reflected from a vibrating surface and the other from a fixed reference mirror—recombine, their relative phase encodes the surface’s motion as a measurable intensity modulation at the detector. In a Michelson configuration, this interference pattern can be expressed as

$$I(t) = I_0 \left[1 + \cos \left(\frac{4\pi}{\lambda} \Delta z(t) \right) \right],$$

where I_0 is the mean intensity, λ is the wavelength of the laser, and $\Delta z(t)$ is the instantaneous displacement of the reflective surface. Variations in the surface position therefore produce time-varying intensity fringes, whose temporal oscillations correspond directly to the vibration frequency of the surface, which has been shown by Fu et al [11]. These principles have long underpinned traditional laser vibrometry systems, which are capable of resolving extremely small displacements by tracking the phase evolution of interference fringes.

However, conventional interferometric vibrometers typically rely on frame-based cameras, which capture sequences of images at fixed frame rates. The temporal resolution of such systems is limited by the maximum

frame rate and exposure time, leading to a trade-off between measurement accuracy, data volume, and computational cost [8]. Capturing fast or high-frequency vibrations requires frame rates that are both technologically demanding and data-intensive, constraining real-time or continuous monitoring applications. Event-based cameras, by contrast, offer a fundamentally different sensing paradigm. Instead of recording full image frames, these neuromorphic sensors asynchronously detect changes in logarithmic intensity at the pixel level. Each pixel triggers an ON or OFF event when the change in log-intensity $\Delta \log I$ exceeds a contrast threshold C , such that

$$\Delta \log I > C.$$

This design allows for microsecond-level temporal precision, high dynamic range, and dramatically reduced data throughput compared to traditional sensors [10, 12]. When integrated into an interferometric setup, the dynamic interference fringes generated by a vibrating surface translate directly into asynchronous streams of ON and OFF events that trace the temporal evolution of the optical phase. In this context, the temporal density of events reflects the frequency and amplitude of the underlying vibration. Rapid phase modulations in the interference pattern produce higher event rates, while slower oscillations yield more sparsely distributed activity. By aggregating these events over time to reconstruct an approximate signal $s(t)$, it becomes possible to analyze the temporal frequency content of the surface motion using Fast Fourier Transform (FFT) techniques. This process effectively bridges optical interferometry and event-based sensing: surface motion is transduced into phase variations, encoded as intensity modulations, and finally discretized into event streams from which vibration frequencies can be recovered.

The theoretical basis presented here establishes the foundation for the subsequent mathematical and analytical framework. The following section formalizes this relationship between optical phase variation and event generation, providing the groundwork for the experimental design and subsequent analyses of event-based interferometric signals.

III. Pixel Behavior and Sensor Noise Classification

A. Expected Pixel Response under Laser Modulation

In an ideal interferometric measurement, each pixel of the event-based sensor receives optical input modulated by the periodic interference pattern formed by the vibrating reflective surface. This periodic pattern can manifest as either temporal intensity modulation at a fixed pixel (when the interference fringes remain stationary) or as spatial fringe motion across the pixel array (when the fringes shift over time) [14]. Under sinusoidal surface vibration, the optical intensity at each pixel follows a corresponding sinusoidal modulation in time, resulting in temporally periodic transitions between increasing and decreasing intensity.

Event generation in the sensor is governed by the relative change in the logarithmic light intensity exceeding the camera’s contrast threshold C_{th} :

$$\Delta \ln I(t) = \pm C_{\text{th}}.$$

This phase-locking mechanism ensures that ON events are triggered near the upward crossings of the modulation waveform, while OFF events occur near the downward crossings. Consequently, in noise-free operation, the timing and polarity of events exhibit strong phase coherence with the driving modulation signal.

Each pixel should therefore produce:

- A near-symmetric count of ON and OFF events over each modulation cycle.
- Phase-locked event generation with minimal jitter across cycles.
- Spatially coherent phase relationships across neighboring pixels within the same interference fringe.

Under ideal conditions, ON and OFF events occur symmetrically in time, mirroring the sinusoidal modulation. However, small deviations—such as unequal thresholds, static illumination bias, or nonlinear pixel response—can lead to asymmetries in event counts or shifts in phase timing. Furthermore, if the optical fringe spacing approaches the pixel pitch, phase aliasing may arise, breaking spatial coherence and producing local inconsistencies in event timing patterns [13]. Deviations from these expected behaviors—such as excess OFF events during bright phases, missing ON transitions, or asynchronous firing—serve as indicators of sensor or environmental noise.

B. Classification of Sensor Noise Mechanisms

Several types of sensor noise can degrade the measurement accuracy and coherence of event-based interferometric data. Under the controlled illumination and vibration conditions used in this experiment, three primary noise categories are identified:

1. **Temporal Mismatch Noise** – Arises when event polarity or timing deviates from the expected modulation phase. Examples include ON events recorded during the decreasing phase of the sinusoid or OFF events during the increasing phase. Such mismatches result from electronic timing jitter, contrast threshold drift, or latency variations within the event-generation circuitry. Temporal noise degrades the phase fidelity of the recorded event stream, causing phase blur and distortion in frequency-domain analyses of the modulation signal.
2. **Spatial Defect Noise** – Refers to pixels that produce inconsistent or unstable responses relative to their spatial neighbors under identical illumination. Manifestations include isolated hot or dead pixels, regions with abnormally high or low event densities, or irregular patterns unrelated to the interference fringes. Spatial noise is typically linked to pixel-level bias variations, fixed-pattern noise, or photodiode non-uniformities. Some defects can fluctuate over time due to thermal drift or local bias instability, introducing temporal as well as spatial variability.
3. **Dark or Spurious Event Noise** – Corresponds to events generated in the absence of optical modulation, occurring under static illumination or DC bias conditions. These spurious events originate from leakage currents, thermal fluctuations, or random electronic noise within the pixel circuitry. Although generally low in rate, such background activity contributes to baseline noise and can bias SNR or coherence estimations if not properly filtered or subtracted.

By quantifying these noise types through temporal mismatch rates and spatial variability metrics, the analysis aims to separate systematic modulation-driven activity from random or pixel-specific noise contributions. This classification provides a foundation for interpreting the temporal and spatial noise measurements presented in later sections.

IV. Data Collection

Overall Experimental Design

The experimental data were acquired using a Michelson interferometric setup designed to capture acoustic surface vibrations through event-based sensing. The setup consisted of a coherent laser source with a wavelength of $\lambda = 600$ nm, which was split into reference and measurement arms using a beam splitter. The measurement arm was directed toward a reflective surface composed of a stretched latex glove, which served as the vibration medium. Acoustic excitation was provided by a calibrated speaker positioned near the latex, driven at controlled sinusoidal frequencies within the audible range (Fig. 2).

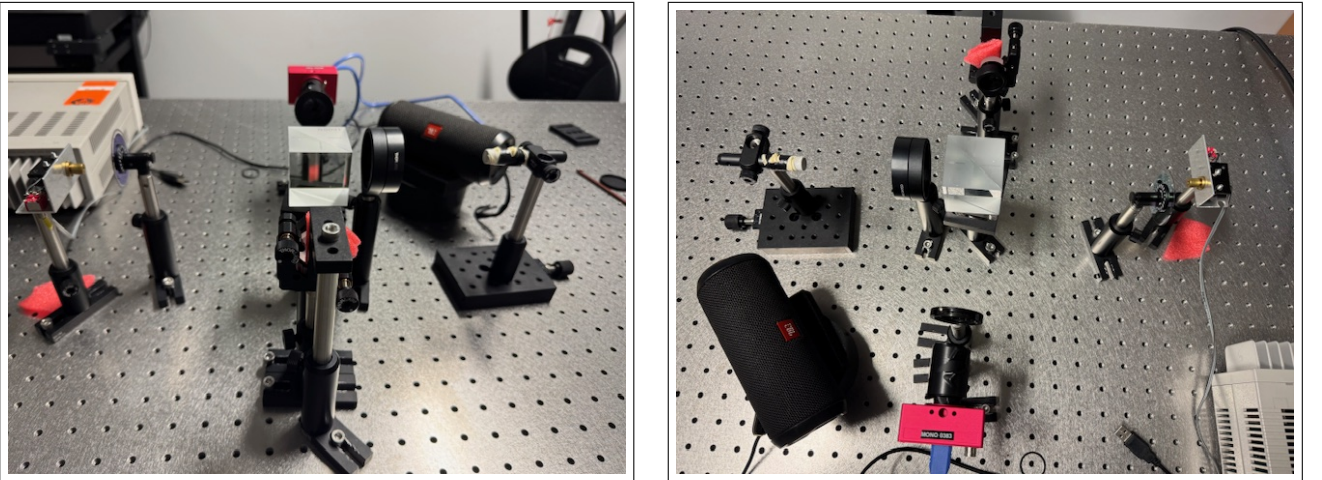


Figure 2: Two viewpoints of the experimental interferometry setup showing the laser, latex surface, and DAVIS346 sensor. Photographs were taken under normal lighting without the cardboard box for clarity.

The surface vibrations induced by the acoustic pressure modulated the optical path length of the reflected laser beam, producing interference fringes when recombined with the reference beam. These interference patterns were recorded using a *DAVIS346* event-based camera, which asynchronously detects changes in light intensity at the pixel level. Unlike conventional frame-based cameras, the DAVIS sensor outputs events corresponding to brightness changes over time, enabling microsecond-level temporal resolution and high dynamic range.

Each experimental trial corresponded to a specific modulation frequency emitted by the speaker. For each frequency, event data were recorded for several seconds to ensure sufficient sampling of the steady-state vibration pattern. In addition, to minimize external disturbances, a cardboard box was placed over the sensor and laser beams, and all measurements were performed in a dark and silent environment. This ensured that only vibrations of the latex surface induced by the controlled acoustic signal were recorded. The resulting event streams were stored as text-based files containing timestamped events of the form (t, x, y, p) , where t denotes the event timestamp (in microseconds), (x, y) are the pixel coordinates, and p encoded as binary values, with 1 representing ON events and 0 representing OFF events. Furthermore, baseline measurements were acquired under two static conditions: (1) with the laser pointed at the center of the latex glove in the absence of speaker excitation, and (2) with the laser directed at the edge of the metallic apparatus to verify negligible mechanical vibrations.

Equipment Parameters

The optical path lengths and key distances of the interferometric setup were as follows:

Component	Distance
Beam splitter $\rightarrow$ cube center	10 cm
Lens $\rightarrow$ box	39.5 cm
Lens $\rightarrow$ cube center	5 cm
Lens $\rightarrow$ camera	9 cm
Camera $\rightarrow$ optical filter	15.5 cm
Filter $\rightarrow$ cube center	9.5 cm
Camera $\rightarrow$ cube center	5.5 cm
Box $\rightarrow$ far mirror	34 cm
Far mirror $\rightarrow$ lens	74 cm

Table 1: Key distances in the event-based interferometry setup.

These distances ensured proper alignment of the reference and measurement beams, as well as optimal focusing on the reflective latex surface. Together, the surface vibrations induced by the acoustic pressure modulated the optical path length of the reflected beam, producing interference fringes when recombined with the reference beam. Acoustic excitation was applied using a calibrated speaker positioned at a set distance from the latex surface. For each frequency, the speaker amplitude was controlled at a constant sound pressure level to ensure repeatable excitation. Each experimental trial corresponded to a specific modulation frequency emitted by the speaker.

Trials and Repetitions

For each frequency, two independent trials were recorded, each lasting approximately five seconds to ensure adequate sampling of the vibrational patterns. A single latex glove was used for all measurements to maintain consistency, and the experimental setup remained fixed between trials. The latex glove was chosen as the vibratory surface because it responded primarily to lower-frequency acoustic inputs. To systematically characterize the sensor response, a frequency sweep was performed across nine acoustic frequencies: 75, 80, 85, 90, 95, 100, 105, 110, and 115 Hz. This procedure enabled comprehensive capture of both the temporal and spatial dynamics of event activity across the relevant vibrational range.

V. Data Processing and Characterization

0.1 Raw Events to Event Accumulation

Prior to any accumulation or processing, the recorded data consisted of raw neuromorphic events directly output by the *DAVIS346* event-based camera. Each event represents a localized change in light intensity at a specific pixel and timestamp, encoded in the form (t, x, y, p) . Figure 3 illustrates a sample of raw events captured during a single experimental trial. In this visualization, individual events are plotted, but due to the asynchronous and

sparse nature of event-based sensing, only pixels experiencing brightness changes generate events. However, raw event streams are inherently noisy, with spurious events arising from sensor noise, background light fluctuations, or minor uncontrolled vibrations. This noise can obscure subtle patterns in the interference fringes and makes direct interpretation of the raw events challenging. Despite this, these raw event streams serve as the fundamental data for all subsequent analysis, including accumulation, filtering, and Fourier-based transformations, where noise can be mitigated and meaningful patterns extracted.

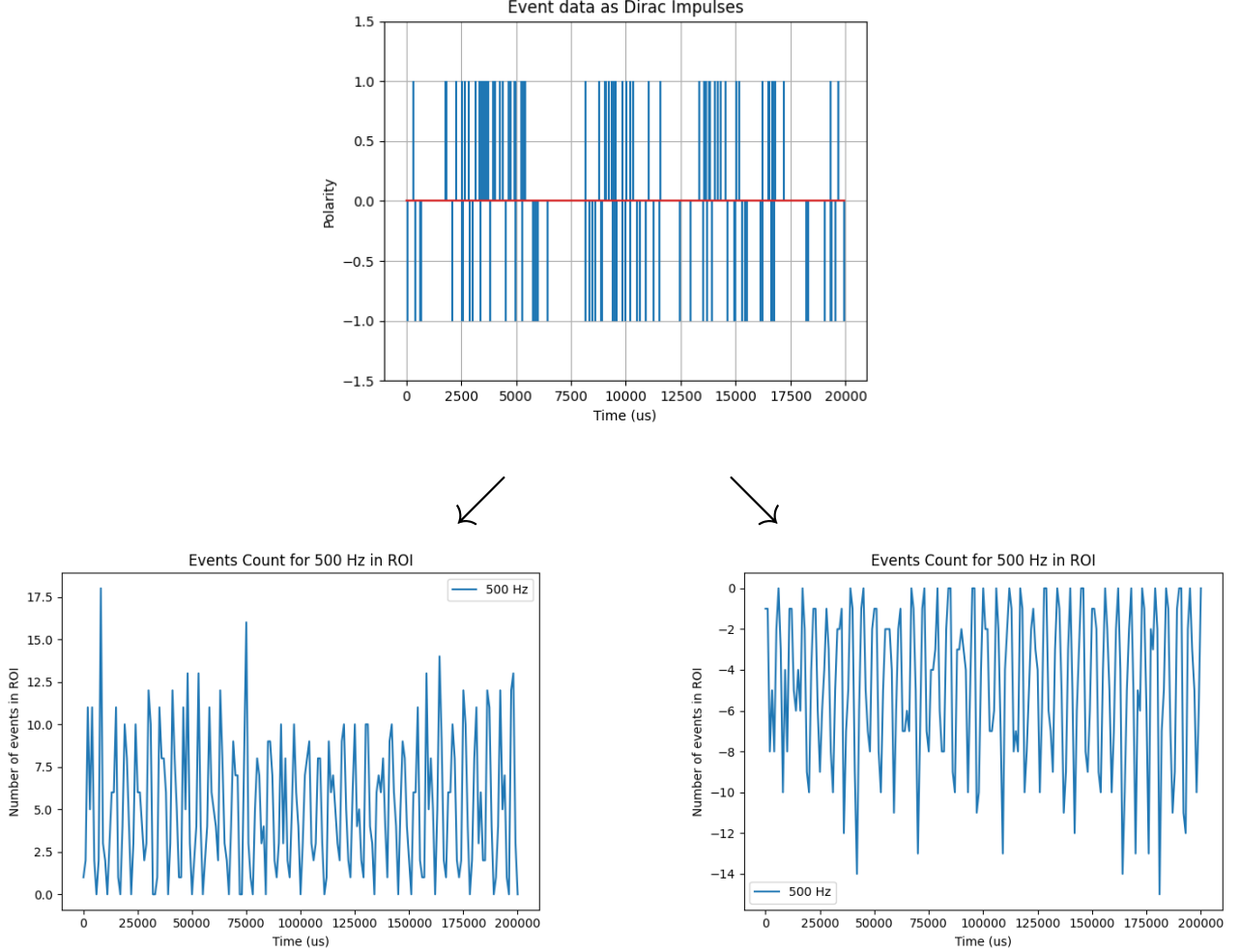


Figure 3: Raw event spikes subsequently accumulated and separated into ON (left) and OFF (right) events.

Thus, event pre-processing began by accumulating ON, OFF, and ON + OFF events. However, the initial accumulation revealed significant noise in the global environment, which prompted the implementation of both spatial and temporal windowing techniques. Spatial windowing helped focus analysis on specific regions of interest, minimizing the influence of peripheral noise, while temporal windowing ensured that only relevant event windows were considered, reducing artifacts caused by transient or irrelevant activity. This approach significantly improved the accuracy of the analysis by reducing noise, eliminating hot pixels, and ensuring that only the most relevant event data were retained for further analysis.

0.2 Spatial Windowing for Event Distributions

To identify regions of maximal event activity within the sensor's field of view, we employed a spatial windowing approach. First, heatmaps of the ON, OFF, and ON + OFF events were generated for each frequency (Fig. 4 shows all heatmaps for 90 Hz). In these heatmaps, "ON" and "OFF" events refer to specific polarities of the events, while "ON + OFF" combines both. Next, a small square window of 10 x 10 pixels was systematically moved across the entire event heatmap, and the total number of ON, OFF, and ON + OFF events within each window position was computed. The window location with the highest accumulated event count was selected as the region of interest (ROI), capturing the area of the sensor with the most significant activity. This window was then zoomed in for visualization, providing a clear depiction of the localized spatial distribution of events.

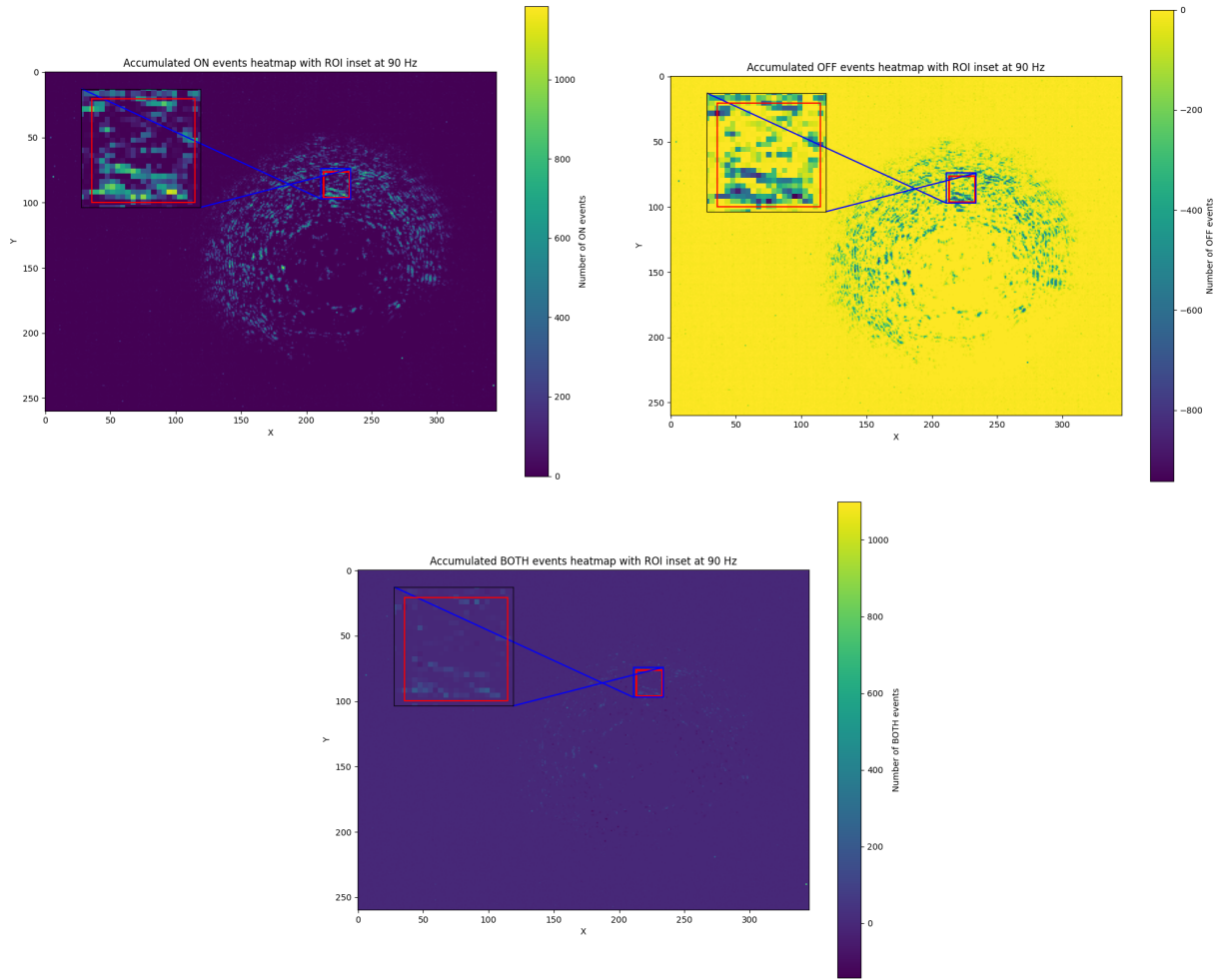


Figure 4: *Zoomed in 10 x 10 spatial window of maximum event activity within the ROI.*

By focusing on the ROI, this method ensures that subsequent temporal analyses prioritize the portion of the sensor exhibiting the strongest response to the stimulus. This approach was applied to all frequency data files, ensuring consistency across analyses.

0.3 Temporal Windowing for Event Distributions

To characterize time-dependent event behavior and quantify temporal noise, a temporal windowing approach was applied to the event stream. For each selected region of interest (ROI), event timestamps were first normalized relative to the start of the recording, and the modulation frequency was used to convert time into fractional phase over one modulation period. The full sequence was then partitioned into discrete temporal bins that subdivided each period into 36 equal phase intervals, enabling event alignment with the expected sinusoidal modulation cycle. Within each window, histograms of ON and OFF events were computed to capture the polarity-phase relationship at fine temporal resolution. These sliding temporal windows provided localized measurements of event coherence, allowing identification of time segments where polarity transitions deviated from the nominal waveform. The approach thus facilitated quantitative analysis of temporal mismatch events—instances where OFF events occurred during rising intensity phases or ON events during falling phases—forming the basis for the temporal noise metrics summarized in the following results section.

0.4 Pre-Processing Metrics for Subsequent Noise Analysis

The table below (Table 2) summarizes ROI-level event statistics and reliability classifications across different modulation frequencies. Two complementary metrics were computed to assess the quality and reliability of ROI activity. First, the total number of events within each ROI and their separation into ON and OFF events quantify how focused the sensor’s responses are within each ROI. Higher event counts concentrated in the ROI indicate better sensor performance and more meaningful responses, whereas low counts or dispersed events suggest background noise or poor sensor alignment. Second, temporal consistency was measured as the stan-

dard deviation of inter-event intervals per pixel within each ROI. Pixels firing very regularly have low standard deviations, indicating consistent responses to the stimulus, while high variability corresponds to irregular firing and increased noise.

ROIs are then classified as Reliable if their variability falls below a threshold (mean + 1 standard deviation) and Noisy otherwise, with ON and OFF events evaluated independently. The choice of threshold was intentional as the traditional mean + 2 standard deviations for denoting significance resulted in errors and all files were found to be Reliable. All ROIs for every modulation frequencies tested (75-115 Hz) were found to be Reliable, whereas baseline and static recordings were found to be Noisy, confirming that the sensor correctly registers low activity under static conditions. These preliminary metrics form an appropriate pre-screening layer for subsequent spatial and temporal noise analyses, ensuring that only regions exhibiting consistent, high-density modulation responses are included in deeper phase-based evaluations. Overall, these pre-processing results demonstrate that the sensor produces robust, frequency-dependent responses in modulated conditions while maintaining low activity and variability under static or baseline conditions, providing confidence in the accuracy of the camera and measurement system.

Filename	ROIEvents_All	ROIEvents_ON	ROIEvents_OFF	ROIavg_std_ON	ROIavg_std_OFF	ROIClass_ON	ROIClass_OFF
75Hz.csv	6505	3284	3221	388602	355343	Reliable	Reliable
80Hz(2).csv	22201	11260	10941	287226	299189	Reliable	Reliable
baseline.csv	2839	1983	856	462186	433466	Noisy	Noisy
static.csv	3080	2158	922	508671	554553	Noisy	Noisy
83Hz.csv	35008	18108	16900	156566	160452	Reliable	Reliable
85Hz.csv	20482	10426	10056	235933	251470	Reliable	Reliable
90Hz(2).csv	61873	32195	29678	116979	118979	Reliable	Reliable
95Hz.csv	49705	26004	23701	145254	153523	Reliable	Reliable
100Hz.csv	12959	6579	6380	392976	336086	Reliable	Reliable
105Hz.csv	9025	4552	4473	305064	311554	Reliable	Reliable
110Hz.csv	5524	2826	2698	374971	346055	Reliable	Reliable
115Hz.csv	18782	9385	9397	306100	330987	Reliable	Reliable

Table 2: ROI-level event statistics and reliability classification across modulation frequencies.

VII. Final Results

Event-based interferometric measurements were processed to evaluate deviations from ideal pixel responses predicted under sinusoidal optical modulation. Three complementary metrics were examined to characterize sensor performance: (i) spatial noise, defined as cross-pixel inconsistency within localized high-activity regions; (ii) temporal noise, defined as per-cycle polarity or phase mismatches relative to the expected modulation waveform; and (iii) the signal-to-noise ratio (SNR), which quantifies the strength of modulation-driven signal components relative to baseline noise. Together, these analyses provide a comprehensive view of both signal quality and stability. All metrics were computed from event streams acquired at modulation frequencies between 75-115 Hz under consistent illumination and vibration amplitude conditions.

Spatial Noise

Methodological Approach

Spatial noise analysis focused on small 10 x 10 pixel windows centered on regions exhibiting maximum ON-event activity. For each dataset, phase histograms were computed by folding event timestamps across the modulation period. Within each region of interest (ROI), an expected polarity profile was constructed by comparing ON and OFF event distributions across 36 uniformly spaced phase bins. Each pixel’s local polarity histogram was compared to this regional profile, and a spatial noise score was defined as:

$$N_s = 1 - \frac{N_{matches}}{N_{total} + 1},$$

where $N_{matches}$ denotes phase-consistent events and N_{total} is the total number of recorded events for that pixel. Pixels exceeding a noise score threshold of 0.5 were classified as abnormal.

Observed Characteristics

Spatial noise values remained tightly clustered between 0.49 and 0.51 across all tested modulation frequencies, indicating near-uniform spatial coherence within the active interferometric regions (Figure 5). The standard deviation per frequency, ranging from approximately 0.02 to 0.05, reveals limited dispersion of pixel responses,

suggesting that only a small subset of pixels exhibited inconsistent polarity-phase behavior. This tight distribution demonstrates that the event-based sensor’s output remains spatially stable under varying modulation conditions.

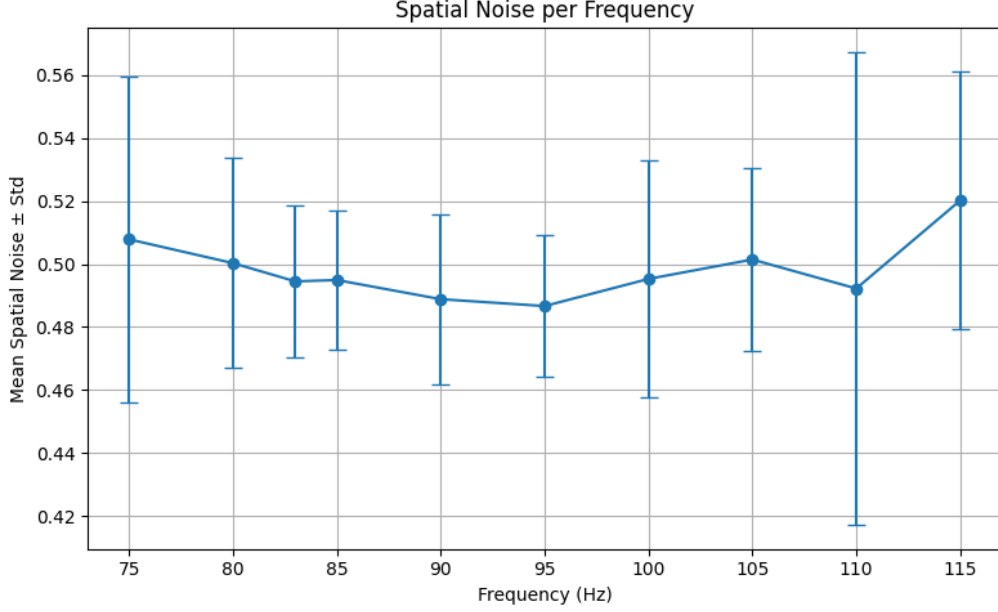


Figure 5: Mean and standard deviation of spatial noise across tested modulation frequencies.

Visual inspection of the spatial noise heatmaps (Figure 6) confirms these statistical observations and further reveals a distinct frequency-dependent trend in noise structure. Heatmaps corresponding to mid-range modulation frequencies (80–95 Hz) exhibited the highest spatial uniformity and the most coherent fringe representation. In this range, low-noise pixels appeared more uniformly distributed and ON/OFF event transitions remained phase-consistent across the ROI. The resulting maps display smooth, well-defined spatial gradients that align with the underlying interference pattern, reflecting optimal synchronization between optical modulation and sensor response dynamics.

At lower frequencies (<80 Hz), the maps appeared more heterogeneous, with a noticeable increase in sparse high-noise pixels. This pattern likely arises from reduced event density within each modulation cycle, resulting in elevated statistical variability and a greater influence of per-pixel threshold bias. Conversely, at higher frequencies (>100 Hz), the noise maps displayed a progressive loss of spatial coherence—characterized by scattered clusters of elevated noise and degraded smoothness. These artifacts are attributed to increased timing jitter, limited pixel bandwidth, and diminished signal-to-noise ratio at rapid modulation rates, consistent with previously observed frequency-dependent degradation in event-camera coherence

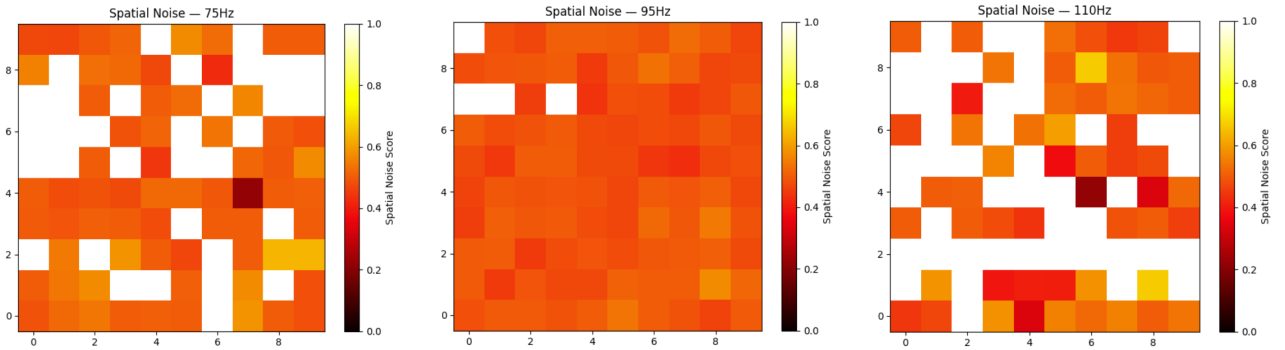


Figure 6: Spatial noise heatmaps of representative frequencies showing increased uniformity in the mid-frequency regime (center) compared to lower and higher frequency extremes.

Interpretation

This observed non-monotonic behavior indicates the presence of an optimal operating frequency band, wherein modulation-induced illumination changes most effectively match the temporal response characteristics of the event sensor. Within this regime, phase-locked event activity produces the most coherent and spatially stable representations of the interference fringes. Frequencies outside this range, either too low or too high, introduce mismatched timing dynamics that accentuate statistical noise or bandwidth limitations. Together, these results demonstrate that spatial noise maps most faithfully capture interferometric coherence in the mid-frequency region, reflecting the sensor’s peak performance window and its inherent temporal constraints.

Temporal Noise

Methodological Approach

Temporal noise quantifies deviations in event polarity and phase alignment relative to the expected sinusoidal modulation cycle. Each event’s fractional phase was computed as:

$$\phi = \frac{t \bmod T}{T},$$

where $T = 1/f$ is the period corresponding to the modulation frequency f . The expected polarity per phase bin was determined using the ROI-level ON/OFF histogram balance. Two complementary metrics were then extracted:

- Weighted mismatch fraction: the ratio of all phase-polarity inversions to total ROI events
- Normalized pixel mismatch rate: the mean per-pixel mismatch fraction within the ROI, normalized by active pixel count.

Observed Characteristics

Temporal noise metrics were evaluated across modulation frequencies ranging from 75 Hz to 115 Hz, using both the weighted mismatch fraction and the normalized pixel mismatch rate to quantify deviations between the observed event polarity sequence and the expected modulation phase. The measured values were largely consistent, with weighted mismatch fractions spanning 0.47–0.50 for dynamic datasets and remaining significantly lower (≈ 0.30) for static or baseline sequences (Table 3). The close agreement between the two metrics confirms that the chosen measures capture the same underlying temporal incoherence, differing primarily in normalization at the pixel level.

Filename	Frequency (Hz)	Weighted Mismatch Fraction	Normalized Pixel Mismatch
75Hz.csv	75	0.495	0.486
80Hz(2).csv	80	0.493	0.487
baseline.csv	80	0.302	0.462
static.csv	80	0.299	0.454
83Hz.csv	83	0.483	0.483
85Hz.csv	85	0.491	0.485
90Hz(2).csv	90	0.480	0.483
95Hz.csv	95	0.477	0.481
100Hz.csv	100	0.492	0.476
105Hz.csv	105	0.496	0.486
110Hz.csv	110	0.488	0.462
115Hz.csv	115	0.500	0.508

Table 3: Temporal noise metrics across modulation frequencies.

For lower frequencies (<80 Hz), the mismatch fraction increased mildly due to uneven event timing within cycles and greater relative influence of per-pixel threshold variability. At higher frequencies (>100 Hz), mismatch rates rose again, accompanied by visible dispersion in phase histograms and increased instances of polarity inversion (Figure 7). These effects are consistent with sensor latency and analog front-end bandwidth limitations, which reduce phase coherence as the modulation period approaches the pixel response limit.

The reduced mismatch observed in static and baseline recordings (≈ 0.30) confirms that the analysis effectively separates modulation-driven temporal noise from background dark or leakage events. The near-symmetric

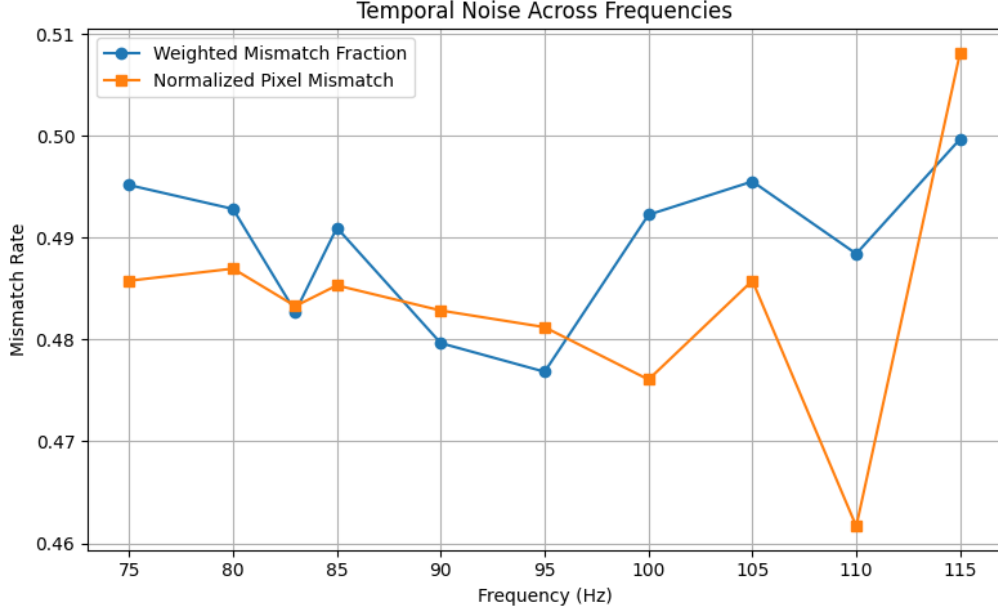


Figure 7: *Weighted mismatch fraction and normalized pixel mismatch across frequencies.*

mismatch behavior across ON and OFF transitions further indicates that temporal noise primarily originates from timing jitter and threshold asymmetry, rather than a systematic polarity bias.

Interpretation

The observed non-monotonic frequency dependence—showing minimal noise in the 80–95 Hz range with moderate elevation at both extremes—demonstrates that the event sensor achieves maximal temporal coherence where the optical modulation dynamics align with the intrinsic sensor response bandwidth. This trend agrees with prior reports that event-camera phase coherence peaks when the modulation frequency closely matches the temporal bandwidth of the pixel front-end circuitry.

Signal to Noise Ratio Estimation

Methodological Approach

To quantify sensor noise, the Signal-to-Noise Ratio (SNR) was estimated from event activity following the approach of Moeys et al., who defined SNR in terms of the variance of stimulus-evoked versus baseline activity [13]. The SNR (in dB) was computed separately for ON and OFF events within the region of interest (ROI). Negative infinite values indicate cases where no measurable signal was detected for a given polarity, typically due to the absence of events. The resulting SNR values reveal a polarity-dependent response, where OFF events show stronger and more consistent signal power compared to ON events, suggesting a possible contrast bias in the recorded data or polarity imbalance in event generation. For each pixel:

$$\text{SNR}_{\text{ON}} [\text{dB}] = 20 \log_{10} \frac{s_{\text{ON}} - n_{\text{ON}}}{n_{\text{ON}}}, \quad \text{SNR}_{\text{OFF}} [\text{dB}] = 20 \log_{10} \frac{s_{\text{OFF}} - n_{\text{OFF}}}{n_{\text{OFF}}} \quad (1)$$

where:

- s_{ON} and s_{OFF} are the measured signal events during the expected ON and OFF cycles, respectively.
- n_{ON} and n_{OFF} are the noise baselines, obtained from measurements when no acoustic stimulation occurs.

This method subtracts the noise baseline from the measured signal before normalizing by the noise, which allows detection of low SNRs. In cases where the measured signal is smaller than twice the noise ($s_{\text{ON}} < 2n_{\text{ON}}$ or $s_{\text{OFF}} < 2n_{\text{OFF}}$), the SNR falls below 0 dB. If the signal equals the noise baseline, the SNR is formally $-\infty$ dB, indicating no input-dependent signal. Since the current experimental setup did not provide a perfectly controlled stimulus, the SNR was inferred by comparing the high-interference region to baseline “quiet” measurements rather than directly measuring an ideal reference signal.

Observed Characteristics

The signal-to-noise ratio (SNR) results presented in Table 4 demonstrate notable polarity- and frequency-dependent variations in modulation coherence within the sensor ROI. Several frequencies show $-\infty$ dB values for either ON or OFF polarities, indicating that, for these cases, the baseline noise counts equal or exceed the modulation-driven signal counts. This outcome suggests that the sensor activity in those polarities and frequency bands is indistinguishable from background noise, reflecting either weak modulation contrast or high inherent noise in those regions. Conversely, positive finite SNR values—ranging from approximately 26 dB up to 58 dB—indicate strong signal dominance over noise, confirming robust modulation event generation in those specific polarity-frequency pairs. Across frequencies, the SNR generally increases with stimulus frequency up to an optimal point before declining, reflecting the sensor’s limited temporal bandwidth and characteristic frequency-dependent response. This trend is consistent with the expected behavior of event-based vision sensors, which exhibit maximal responsiveness within a specific range of temporal contrast changes. However, for certain recordings, noise dominates the signal, leading to lower or undefined SNR values. Such cases likely correspond to either weak stimulus modulation or increased background activity, emphasizing the importance of baseline normalization when interpreting event-driven SNR metrics.

File	Frequency (Hz)	ROI SNR ON (dB)	ROI SNR OFF (dB)
75Hz	75	$-\infty$	26.53
80Hz(2)	80	$-\infty$	44.21
83Hz	83	$-\infty$	55.00
85Hz	85	41.20	$-\infty$
90Hz(2)	90	$-\infty$	57.62
95Hz	95	55.42	$-\infty$
100Hz	100	$-\infty$	45.68
105Hz	105	$-\infty$	31.19
110Hz	110	41.35	$-\infty$
115Hz	115	$-\infty$	44.44

Table 4: Signal-to-noise ratio (SNR) in decibels (dB) for ON and OFF event polarities at various modulation frequencies. The ‘ $-\infty$ ’ values indicate cases where the noise count met or exceeded the signal count, resulting in an undefined or non-measurable SNR. These values signify that no meaningful signal dominance over noise was detected in the ROI for that polarity.

Interpretation

These results highlight the nuanced interplay between sensor polarity response and signal quality in event-based interferometric measurements. Although the total count of ON events exceeds that of OFF events, the stronger signal-to-noise ratios observed for OFF events reflect intrinsic sensor characteristics and the spatiotemporal dynamics of event generation. Polarity-dependent biasing, threshold tuning, and differential contrast sensitivity can cause OFF channels to produce more stable modulation responses in noisy conditions, despite appearing less active overall. Additionally, asymmetries in illumination gradients, scene reflectance, and baseline dark noise may further accentuate these differences. This polarity-specific behavior reinforces the necessity of treating ON and OFF events as independent sensing modalities when analyzing noise and signal coherence in event cameras. The adopted SNR framework, combined with spatial ROI selection, enables precise localization of sensor regions where modulation-driven activity reliably exceeds baseline noise, thereby facilitating robust characterization of interferometric performance across frequency and polarity domains.

VIII. Discussion

The present analysis provides a multifaceted characterization of sensor performance in event-based interferometric measurements, highlighting the interplay between signal modulation, sensor polarity response, and inherent noise mechanisms. By examining spatial noise, temporal consistency, and SNR, several key insights emerge.

First, spatial noise metrics demonstrated high coherence within active ROIs, particularly in the mid-frequency regime (80–95 Hz), reflecting uniform phase-locked responses across pixels. Deviations observed at lower and higher frequencies indicate that pixel-level nonuniformities, event sparsity, and bandwidth limitations manifest as localized noise clusters. These findings emphasize the importance of ROI selection in robustly quantifying sensor performance and suggest that optimal spatial coherence occurs when modulation dynamics align with

the sensor’s intrinsic spatial and temporal response characteristics.

Temporal noise analysis showed minimal mismatch fractions in the same mid-frequency band, indicating strong phase coherence of event generation. Increased mismatch at low and high frequencies correlates with slower modulation cycles, limited event counts, or front-end bandwidth constraints. Temporal noise in static and baseline recordings was markedly lower, confirming that the metrics effectively distinguish between modulation-induced temporal variations and background dark or leakage events.

Finally, SNR analysis revealed a non-monotonic frequency dependence, with peak signal dominance occurring within a mid-range modulation window. OFF-polarity events often exhibited higher SNR values than ON-polarity events, despite fewer raw counts, suggesting that sensor biasing, threshold settings, and intrinsic pixel response dynamics differentially affect polarity channels. The presence of $-\infty$ dB values in several files highlights conditions where the baseline noise overwhelms the modulation signal, providing a clear indicator of frequency-dependent limits in detectable modulation contrast.

Several limitations constrain the generality of these findings. First, the lack of a perfectly controlled reference signal prevents precise quantification of absolute SNR, requiring inference relative to baseline measurements. Second, the analysis assumes quasi-uniform modulation within ROIs, which may not fully capture local inhomogeneities or optical interference effects outside selected windows. Third, the high baseline noise in some datasets limits the interpretability of SNR measurements, especially for ON-polarity events at certain frequencies. Finally, the current metrics do not fully disentangle optical, acoustic, and electronic noise sources, meaning the results primarily reflect sensor-intrinsic noise superimposed on experimental variability.

Future investigations could extend this analysis by incorporating more controlled reference measurements to better isolate modulation-induced events from baseline noise. High-speed imaging or synchronized photodiodes could provide ground-truth temporal signals, enabling calibration of temporal noise metrics. Spatially resolved characterization across larger sensor areas could identify systematic nonuniformities, informing sensor design or biasing strategies. Additionally, exploring adaptive ROI selection, frequency-dependent threshold tuning, and cross-polarity correlation analysis could enhance the interpretability and robustness of event-based interferometric measurements, guiding the development of optimized neuromorphic sensing systems for high-fidelity optical measurements.

IX. Conclusion

The combined SNR, spatial, and temporal noise analyses collectively demonstrate that the event-based sensor exhibits distinct frequency-dependent coherence characteristics, with an optimal modulation range emerging between approximately 80 and 95 Hz. Within this regime, both spatial and temporal noise metrics reached their minima, and the signal-to-noise ratio peaked, indicating synchronized alignment between optical modulation and the intrinsic temporal bandwidth of the pixel circuitry. Frequencies outside this band revealed divergent degradation mechanisms: low-frequency operation produced sparse, statistically variable event patterns, while high-frequency modulation introduced temporal jitter and loss of spatial coherence. Polarity-specific analysis further highlighted that OFF events, although less numerous, provided more reliable modulation encoding with higher effective SNRs. These findings establish a comprehensive understanding of the sensor’s spatiotemporal response behavior, revealing how polarity asymmetry, pixel bandwidth, and noise statistics collectively define performance limits in dynamic interferometric measurements. The results provide a foundation for refining event-based sensing models and optimizing future experiments through frequency tuning, polarity calibration, and adaptive noise suppression.

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